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# Decoding the Algorithm: The Mathematics Behind TikTok's Short-Form Content Success

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# Decoding the Algorithm

## The Mathematics Behind TikTok's Short-Form Content Success

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# Abstract

Within the realm of social networks, TikTok has become the central hub for short-form video content. The network's unique ability to capture individual preferences using predictive analytics has greatly contributed to its massive success, allowing the company to optimize its performance and content personalization. In an age where digital media have such a significant influence on society, it is essential that users develop an understanding of how social network algorithms function to make more informed online decisions. Although *TikTok*'s technological system is primarily undisclosed, the platform certifiably leverages several key mathematical principles within its algorithm to achieve its core goals of generating user, creator, and platform value. This thesis reveals how mathematical modeling is embedded into the application's design in order to optimize content delivery, personalize the user experience, and ultimately increase engagement and platform growth. Matrix factorization is implemented in the application's operating system to track video-user interactions and predict engagement. The network uses probability models to estimate the likelihood that a user will engage with a video based on past behavior, observing metrics such as watch time, likes, and shares. With the help of artificial intelligence, machine learning algorithms analyze videos to match user interests and adjust recommendations in real time. Lastly, the use of Fourier transforms allows the company to perform trend analysis on sound waves, predicting the success of music and speech on the platform. Each component of the algorithm ensures that the platform produces high-quality content recommendations that encourage long-term engagement from users.

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# 1 Introduction

## 1.1 Background

The explosion of the viral short-form video application, *TikTok*, has completely reshaped the trajectory of social media and content consumption. With over 1.5 billion active users worldwide, the app's unique features and "addictive" algorithm help engage long-term viewership and set the service apart from its competitors.

The emergence of *TikTok* has a long-standing history of mergers and company improvements. In April 2014, Shanghai tech entrepreneurs Alex Zhu and Luyu Yang released an application prototype called *Musical.ly*. Motivated by the microcontent trend facilitated by *Vine*, (a popular loop video host), the new lip syncing platform allowed users to create 15-second to 1-minute videos set directly to music soundtracks. By 2016, the app amassed 90 million users worldwide, gaining extreme traction in the United States. Around the same time, Chinese tech powerhouse, Bytedance, launched a similar platform called *Douyin*, which became wildly platform in Thailand and China due to its built-in editing tools and vertical video orientation. With the goal of expanding *Douyin* across the world, ByteDance acquired *Musical.ly* in 2017 and completely rebuilt the platform [17]. The newly transformed social media outlet - which we now know as *TikTok* - launched in 2018 and almost immediately reached number one on the charts in more than 13 countries [14].

By 2019, *TikTok* became the most downloaded app worldwide, completely surpassing *Instagram* and *Facebook* — which for many years remained at the top of the social media hierarchy. The platform is constantly evolving; while it started out as a hot spot for music and dance related content, it quickly grew into a versatile hub for all genres of short-form content. The platform has truly made a cultural impact. Many of the trends that originate on *TikTok* eventually boil over into the outside world, creating a kind of phenomenon surrounding content creation and taking it to completely new heights.

*TikTok* has redefined music discovery for audiences, providing a space for artists to launch

their projects to mass popularity. The platform is notorious for driving music up the charts, as many top "sounds" on the app eventually emerge as hits on top audio streaming platforms. Since the evolution of *TikTok*, the music industry has experienced a drastic change. Artists now use the platform to reach a wider audience and create engaging content surrounding their songs to gain traction. This "influencer marketing" tactic helps increase their exposure and overall revenue generation while simultaneously fostering an online fan community.

The app has also reinvented what it means to be a celebrity, creating a new generation of content creation where virtually anyone can amass a following and become "influential" overnight. Other platforms like *Instagram* and *Pinterest* are known for their polished content, but *TikTok* emphasizes the importance of being raw and relatable. Users feel more connected to the creators that they engage with, seeing more of their true personalities, and getting an inside look into what goes on behind the scenes. This inspires individuals to create on the platform, increasing the amount of content available and appealing to a wide range of user demographics.

*TikTok* has even had an extreme impact on the participation of the younger generation in social and political change. The platform has become a powerful space for advocacy and activism, using short-form content to engage and educate viewers on important issues.

Most importantly, *TikTok* is accessible to anyone, anywhere. As long as you have a mobile device, you can create content. This component of accessibility has broadened the app's global reach, lowering the barrier of entry for content creation and connecting users worldwide.

With *TikTok* being such a controversial part of culture in today's climate, it is integral that users gain a deep understanding of how the app functions. While the system is quite complex and proprietary, there are several clear components to the platform's operation that, when thoroughly comprehended, can enhance the overall experience for both audiences and content creators.

## 1.2 Overview of User Interface

The key characteristic of the app is a "For You" page, which directly streamlines content to each individual according to their interests. This personalization aspect really separates *TikTok* from other social networks and has left many wondering how exactly user preferences are gathered and organized by the software. Although the exact algorithm has not been publicly revealed, a leaked document produced by TikTok's engineering team provides a few key insights about the makeup of the recommendation system at the core of the platform.

*TikTok*'s current Chief Executive Officer, Shou Zi Chew, defines the mission of the app to be to "inspire creativity and create joy" [15]. The goal is to empower users to freely express themselves while fostering an entertaining online environment. Within the company's business model are four main pillars for the performance success of the algorithm: short-term user value, long-term user value, creator value, and platform value.

The value of the user signifies the immediate satisfaction of individuals and the relevance of the content they are fed. The algorithm is built to deliver videos that resonate with current moods and preferences to be rewarded for instant engagement in return.

Since users' preferences are ever-evolving, the platform also takes into account long-term user value. In order to keep individuals engaged in the long run, the model is flexible, predicting changes in needs and behavior in real time.

Since the platform relies on user generated content, TikTok strives to incentivize creators to continue producing the high quality content that drives their engagement. Creator value focuses on filtering a steady stream of diverse and appealing media while also providing users with valuable in-app features for video production.

Lastly, TikTok places emphasis on platform value, which is characterized by growth and stability. The company constantly tweaks and reformulates the algorithm to ensure that the app performs optimally and is easy to use.

Beyond this, the document briefly dives into the two main metrics that are tracked to aid in the generation of daily active viewership. The first is retention, which distinguishes whether

or not a user continues to watch a particular creator, trend, etc. The second characteristic is watch time, which accounts for the duration an individual stays on a given video. How long a user views a video for is a key indicator of their interest; if they swipe past it quickly, the algorithm assumes that the individual does not like that type of content. On top of these factors, *TikTok* records user interactions to form predictions on user preferences. Using a prediction driven by machine learning, user behavior is summed for the following data points: likes, comments, follows and shares. Based on these summations, the system scores each video and generates the content with the highest score to each user.

While the document outlines the broader components of the algorithm, the technical methodology behind the app's operation remains to be publicly outlined by the company. However, using these revealed details and an applied understanding of mathematics, we can develop a deep understanding of how *TikTok* achieves its value pillars through three main components: content recommendation, content personalization, and content production assistance. This thesis will dive into the mathematical functions at the core of these in-app features and how they work in tandem to enhance user experience, empower creators, and support overall platform satisfaction. The remainder of the thesis goes as follows. In Section 2, I will focus on the generation of user value through content recommendation. In Section 3, I will pivot to platform value creation, concentrating on content moderation and personalization. In Section 4, I will touch on creator value and the use of content creation tools on the app. Lastly, I will conclude with the impact of *TikTok*'s algorithm and the potential negative effects that it poses on users.

# 2 Creating User Value through Content Recommendation

This section explores how TikTok's content recommendation algorithm creates value for creators through matrix factorization and feedback loops. By representing users and videos as vectors in a latent space, matrix factorization helps the algorithm predict which content will resonate with users. As users engage with videos, the matrices are updated with interaction data, forming a feedback loop that continuously refines recommendations. This not only boosts user retention but also amplifies creator visibility by matching content with receptive audiences, effectively rewarding creators whose content aligns with engagement patterns.

## 2.1 Matrix Factorization for Data Decomposition

Content recommendation is arguably the driving force behind *TikTok*'s success. Unlike other social media platforms, the app does not sort content chronologically, but rather employs linear algebraic principles within its algorithm to predict what videos users are most likely to engage with. In real time, the platform organizes large amounts of data compiled from users and efficiently ranks content based on potential interest.

Perhaps the most integral mathematical component of TikTok's recommendation system is its utilization of large matrices to assemble information regarding user-video interactions. Using matrix factorization and singular value decomposition (SVD), the system can provide a viewer with video suggestions that are similar to their previous preferences.

The user-video matrix is comprised of rows that represent each individual user and columns that designate pieces of media. The value in each cell evaluates the interaction based on feedback, in the form of likes, comments, shares, etc. There are two ways this matrix can be refined, the first being through the content filtering approach. This strategy creates a profile for each user, characterizing their interest patterns and overall taste. The other approach, which is much more nuanced, is the collaborative filtering approach. This technique has the ability to discover other users with similar preferences and use their ratings

to predict whether or not someone would enjoy a given piece of content. The benefit of collaborative filtering is that it is entirely domain-free and has the capacity to target data that is typically more difficult to profile. Thus, the recommendations produced by the process are significantly more accurate. However, the downside is that it is initially difficult to get this up and running. Collaborative filtering requires preliminary behavior data in order to operate, so it has difficulty addressing the system's new users.

Collaborative filtering consists of two main procedures, the first being neighborhood methods. These are centered around relationships between users, taking into account each user's preference to a video based on the ratings of neighboring (related) videos. The other practice is the use of latent factor models, which characterize each video and user based on various factors for rating patterns. These models measure how much each user likes a given video that has a high correspondence to such a factor. For instance, the system may push a piece of gym-related content to the viewer to estimate the individual's potential interest in fitness. Implementing this into matrix factorization, videos and items are characterized by vectors of factors inferred from previous video ratings. High correspondence to a factor leads to a recommendation. This approach is extremely flexible and highly scalable, which contributes to its precise predictive accuracy.

The most convenient data for a recommendation system to work with is explicit feedback: likes, shares, comments — measurable values. The benefit of leveraging matrix factorization within an algorithm, though, is that it allows for the incorporation of implicit feedback. To observe user behavior, the system may take into account a user's search patterns to make better predictions. Here is a technical description of the mathematics behind the process of forming a recommendation model:

**Algorithm Behind Recommendation Model**

*Proof.* Users and videos are mapped to a joint latent factor space of dimension $f$.

Let $i$ represent videos, with each $i$ associated with a corresponding vector $q_i \in \mathbb{R}$

Let $u$ represent users, with each $u$ associated with a corresponding vector $p_u \in \mathbb{R}$.

For each $i$, the elements in $q_i$ represent the extent to which each video possesses a particular factor. For each $u$, the elements in $p_u$ measure the extent of interest the particular user has in videos that align with the corresponding factors.

We can take the dot product, ${q_i}^T \cdot p_u$, which represents the interaction between each user and each video (i.e. the overall interest the user displays in the video's characteristics). We label this dot product $r_u i$ to represent the approximate rating of the video.

By mapping each video and user to factor vectors $q_i$ and $p_u$, the algorithm can estimate the rating a user will give any given video. Performing singular value decomposition, latent factors are identified in information retrieval. In order to factor the user-video rating matrix, though, we must make the matrix more dense since there are many missing values. Filling these holes with additional data can be quite expensive, so companies typically opt to fit previously observed ratings to help the algorithm learn the model. We can minimize the regularized squared error on the set of known ratings:

$$\min \sum (r_u i - {q_i}^T p_u)^2 + \lambda(\|q_i\|^2 + \|p_u\|^2). \tag{1}$$

Using this method avoids potential overfitting of the observed data by regularizing the learned parameters. From here, there are two ways to minimize the equation.

(i) Stochastic Gradient Descent:

The system predicts $r_{ui}$ for each case and computes the prediction error, $e_{ui}$:

$$e_{ui} = r_{ui} - {q_i}^T p_u. \tag{2}$$

Then, the system modifies the parameters by a magnitude of $\gamma$ in the opposite direction of the gradient.

$$q_i \leftarrow q_i + \gamma(e_{ui} \cdot p_u - \lambda \cdot q_i). \tag{3}$$

$$p_u \leftarrow p_u + \gamma(e_{ui} \cdot q_i - \lambda \cdot p_u). \tag{4}$$

This technique is quite easy to implement and runs fairly quickly, which makes it attractive to use.

(ii) Alternating Least Squares:

In cases where we don't know both $q_i$ and $p_u$, we can fix one of the unknowns and solve the resulting quadratic equation. The system rotates between fixing the $q$'s and $p$'s, solving using the least squares method and decreasing the equation until it converges. The benefit of this method is that works particularly well with implicit data. It is a much more practical approach and can use parallelization, computing each $p$ and $q$ independently of other factors.

It is important to account for potential biases within the recommendations. We do so through the following computation:

$$b_{ui} = \mu + b_i + b_u. \tag{5}$$

$$\hat{r}_{ui} = \mu + b_i + b_u + q_i^T p_u. \tag{6}$$

$$\min \sum (r_{ui} - \mu - b_u - b_i - p_u^T q_i)^2 + \lambda(\|p_u\|^2 + \|q_i\|^2 + b_u^2 + b_i^2). \tag{7}$$

When users don't provide enough information, the recommender system can use implicit feedback to gain additional insight on user preferences. These can come in the form of purchase data, browsing history, etc. We denote this implicit feedback as

$$\sum x_i. \tag{8}$$

The system can also use user attributes and demographics such as age, gender, location, and

income level. We denote this information as

$$\sum y_a \tag{9}$$

Thus, taking this into account, our rating model takes the following form:

$$\hat{r}_{ui} = \mu + b_i + b_u + q_i^T[p_u + \big|N(u)\big|]^{-0.5} \sum x_i + \sum y_a. \tag{10}$$

Since user preferences are ever changing, matrix factorization is helpful in modeling the temporal effects that occur. For instance, video popularity changes over time, or a user's baseline ratings change over time as they find videos they prefer more. In turn, separating ratings into video biases, user biases, and user preferences allows the system to more accurately address each individual change. For dynamic prediction modeling at time $t$, the system uses the following equation [7]:

$$\hat{r}_{ui}(t) = \mu + b_i(t) + b_u(t) + q_i^T p_u(t). \tag{11}$$

□

Matrix factorization is an incredibly helpful tool within the system's algorithm. Its ability to handle sparse data especially comes in handy when it comes to new users who have only interacted with a small set of initial videos. This algebraic technique is able to predict user interactions despite their early experience on the platform, and target the types of content that best appeal to their preferences. The technique can also reveal why users interact with a given piece of content, also known as latent factors [13]. The user matrix contains features that represent different dimensions of user preferences; for instance interest in comedy or fitness. While these have not been precisely outlined in the data, the algorithm infers these interests based on the user's behavior and categorizes videos based on their content genre. The same goes for the video matrix, which distinguishes content based on attributes like

length, energy level, and engagement level. The system learns to train these latent features based on user interactions, and unveils information about a user's affinity to various forms of content. With these capabilities, *TikTok* is able to personalize the overall experience for their users, tailoring the content to the tastes of each individual and ensuring that they continue to come back for more.

## 2.2 Feedback Loops for Engagement Analysis

To further assist in the content recommendation process, the TikTok algorithm likely leverages the cyclical, probabilistic method of feedback looping. Within the context of user interactions, data is collected from each user to understand engagement patterns and individual preferences. From an algorithmic standpoint, this feedback allows for adjustments of content suggestion, prioritizing media that best resonates with users. From the creator side, the information is insightful to content strategy, identifying what performs best on their pages and will get the most attention from the audience. The loop consists of a few distinct components. When a user likes or shares a piece of content, an algorithm updates the user's profile to signal that they enjoy the type of media, which increases the likelihood of them seeing similar content in the future. Completion rate measures the percentage of users that watch a video to its end, which provides a good indicator of interest. The formula is as follows:

$$P_{complete} = \frac{\text{watch time}}{\text{total video length}}. \tag{12}$$

The higher a video's completion rate is, the higher it will be "ranked" on the platform due to the demonstrated user interest. Comments and follows are ranked more heavily in the loop, suggesting deep interest in a given creator/style of media. Lastly, disengagement serves an extremely powerful tool for determining preferences. If a user scrolls past a video or does not interact, it signals to the algorithm that they are not interested, and the platform pushes less of that content their way. For each user, $u$, and video, $v$, an engagement score is calculated

to deliver a feedback signal.

$$\begin{aligned}\text{Engagement Score}(u,v) = (w_1 \cdot \text{Likes}(u,v)) + (w_2 \cdot \text{Comments}(u,v)) \\ + (w_3 \cdot \text{Shares}(u,v)) + (w_4 \cdot \text{Completion Rate}(u,v)).\end{aligned} \tag{13}$$

The variables $w_1$, $w_2$, $w_3$, and $w_4$ represent the weights of each interaction, which are constantly updated for each individual user. Using this information, the system's model is updated instantaneously.

$$\theta_{t+1} = \theta_t + \alpha \times \nabla_\theta \text{Engagement Score}(u,v). \tag{14}$$

$\theta_t$ represents the model parameters at time, $t$, which essentially revolves around the weights of each interaction for the current algorithm. $\alpha$ is the learning rate for the model, controlling how often the algorithm is updated with new information. $\nabla_\theta$Engagement Score$(u,v)$ is the gradient (or rate of change) of engagement score based on the parameters. This determines what the model needs to adjust based on the feedback it received from users. Each user's calculated interactions act as a "reward signal" for the model to update recommendations that better align with maximized performance capabilities and interest.

To obtain suggestions and maximize engagement in real time, feedback loops are also used in a machine learning technique called reinforcement learning. With this technique, the algorithm acts as an "agent" that makes decisions on which videos to recommend to a user on their feed. Based on user's interactions with the content, the system receives a "reward". This strategy is modeled by the Markov Decision Process (MDP), which the following parameters:

- $S_t$: state of the system (user's profile) at time $t$
- $A_t$: action taken at time $t$ (recommendation of a video)

- $R_t$: reward received at time $t$ (user engagement/interactions)
- $\theta_t$: parameters of recommendation model (previous learnings of system)

The objective of this process is to maximize the cumulative reward over time by making constant adjusts to the model for improvement. The future of these rewards can be estimated using a quality function:

$$Q(S_t, A_t) = \mathbb{E}[R_t + \gamma \dot{Q}(S_{t+1}, A_{t+1})]. \tag{15}$$

$Q(S_t, A_t)$ refers to the expected reward for taking action $A_t$ at a given state $S_t$. Also known as the discount factor, $\gamma$ provides the algorithm with control over the value and importance of each reward. $\mathbb{E}[R_t]$ is the immediate expected reward after $A_t$ based on feedback. It directly correlates with the engagement score for a given video. The reinforcement learning model continually updates $\theta_t$ based on the rewards it receives, using an iterative process to calculate $\max_a Q(S_{t+1}, a)$, which is the future rewards the model expects to obtain from the next profile or adjustment.

Using the data collecting from user interactions, *TikTok* creates and updates a user profile for the individual to help personalize their experience and store their engagement patterns. Using feature vectors that represent data numerically, the system is able to compute the probability that a user will engage with a particular video. $U_u$ is the feature vector that represents the user's preferences, obtained from reinforcement learning on their past interactions. $V_v$ is the feature vector that represents the characteristics of a particular piece of content. Taking the dot product of these vectors

$$P_{\text{engagement}}(u, v) = U_u \cdot V_v \tag{16}$$

yields the probability that a user $u$ will engage with a video $v$. If the user interacts with the video, then the system will update the user's profile and recommend similar content in the

future. The system update is calculated as:

$$U_u^{\text{old}+\alpha\cdot(\text{Interaction Feedback})} = U_u^{\text{new}} \tag{17}$$

with "Interaction Feedback" being derived from the reward signals produced earlier by the reinforcement learning model. As more and more feedback is applied, the user's profile becomes much more accurate to their preferences. [6]

Through its sophisticated content recommendation system, *TikTok* generates both short-term and long-term user value, using interaction analysis and dynamic content ranking to identify patterns in engagement and deliver a continually entertaining experience for users.

# 3 Generating Platform Value through Content Moderation and Personalization

This section examines how *TikTok* generates platform value through a combination of AI-driven content moderation and personalized delivery. Using key machine learning models, *TikTok* identifies and filters harmful content, ensuring a safer and more engaging environment for users. AI is used in personalization strategies, such as split testing different content variants and analyzing hashtag performance, to optimize how videos are distributed on the platform. These techniques allow the app to fine-tune its recommendation engine in real time, enhancing user experience and increasing content discovery capabilities.

## 3.1 Video Classification with Artificial Intelligence

Artificial Intelligence (AI) has become a huge buzzword within the digital media space and many social networks have adapted their algorithms with new machine learning technologies to further enrich the user experience of their interface. *TikTok* uses AI in various ways on its platform, the first being in the form of video classification. Using computer vision models and facial recognition software, the app is able to analyze and categorize content based on the objects showcased in the video footage [2].

An essential set of deep learning algorithms for image recognition are Convolutional Neural Networks, which break down pixel data into layers based on patterns. Each image is represented as a matrix, where each element corresponds to a pixel and is classified as a value based on color and intensity. For grayscale images, the matrix is 2D ($I \in \mathbb{R}^{m \times n}$), while for colored images, the matrix is 3D ($I \in \mathbb{R}^{m \times n \times 3}$ for color channels red, green, and blue). Convolution consists of applying a small filter matrix, called a kernel, to each image matrix and performing various mathematical operations to each element to detect the features of the image. A scalar product between the pixel weights and regions is computed, which allows for the calculation of output neurons that are connected to local regions of the image. Let

$K$ represent a kernel of size $k \times k$. Then, the convolution is expressed through the following equation:

$$S(i,j) = \sum_{a=-[k/2]}^{[k/2]} \sum b = -[k/2]^{[k/2]} K(a,b)\dot{I}(i+a, j+b) \tag{18}$$

where $S(i,j)$ is the resulting feature map, which highlights edges, textures, and corners of the image. After the image has undergone convolution, a rectified linear unit (ReLU) is applied to the compiled feature map to identify the non-linear, complex layers of the image that are essential to object recognition.

**Definition 1** (Rectified Linear Unit)**.**

$$ReLU(x) = max(0, x). \tag{19}$$

*If $x > 0$, the value is stored as $x$ If $x \leq 0$, the values is replaced with $0$.*

9m

$$y_{pool}(a,b) = max\{y(p,q)\} \tag{20}$$

where $p, q$ represent the pooling window. The maximum element from the region of the map that is covered by the kernel is selected. The product is a max-pooled layer map representing the most prominent features of the previous mapping. [19]

Following the pooling layer, the neural network creates a "fully connected" layer that flattens the map and connects every input node to every output node. A weighted sum of all of the input is taken to determine feature importance and a non-linear activation function is applied. For a given input vector $y$,

$$y = Wx + b \tag{21}$$

where $y$ represents the output vector. $W$ is the weight matrix of the layer, which discloses the strength of connections between each neuron in the previous layer and the current layer.

Each neuron also has an associated bias value, which assists in shifting the activation function and creating a better fit for the data. This bias vector is represented by $b$. The final fully connected layer, also referred to as the output layer, connects one neuron to each class (feature) within the data set. This layer uses a softmax function, which converts the output scores of the network into a probability distribution over the set of classes. Each probability represents the likelihood that the input feature belongs to a particular class. The softmax of each feature is computed using the following formula:

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}. \tag{22}$$

The variable $p_i$ is the probability of a particular class. $z_i$ is the raw score for each class, determined by the neural network. $e^{z_i}$ ensures that all values are positive, creating confidence in the predictions. And lastly, $\sum_{j=1}^{K} e^{z_j}$ is the normalizing factor that ensures the sum of probabilities equal 1. The softmax function is an integral component to ensuring that the model can automatically extract features, deciding which class the image most likely to belongs to. The complexity of CNNs allow for pattern detection amongst various levels of abstraction, recognizing the following classes [8]:

- Low level features: simple image components like corners and edges
- Mid-level features: more complex textures and contours like eyes and wheels
- High-level features: abstract features like faces and animals.

There are a variety of ways with which *TikTok* likely leverages image CNNs within their algorithm; the first being for content moderation. Safety is a primary concern for the company, with there being a firm set of Community Guidelines that outline what is and is not allowed on the platform. Having these rules in place ensures that users continue to have a welcoming and entertaining experience on the platform. Image recognition technology allows the app to identify and remove any harmful content that goes against these guidelines. When a video is

uploaded to the platform, it is processed through several layers of the CNN to identify any potentially explicit content. The four most prominent features that are scanned for are nudity, violent/graphic behavior, offensive imagery, and explicit gestures. The video is broken down into frames per second, and each frame is resized to be properly suited for the CNN. Then, the frames are passed through several convolutional filters, that each aim to capture a specific category of features mentioned above. Once the data is passed through the fully connected layers, each frame is given an output probability score that classifies it as either "explicit" or "safe." If the probability exceeds a certain threshold, the content is flagged or removed. If the content is deemed safe, then it gets filtered through and posted.

CNN are also a driving force behind content recommendation on an individual's FYP. The network assesses the objects and actions within a particular video to match it according to the user's interests. This ability helps drive engagement on the platform. For instance, if an individual enjoys sports related content, the algorithm will scan for sports-related objects and push any videos that have that connection. This technique is particularly responsible for trend amplification, detecting visual elements that are popular on the platform and continuing to intensify their reach in order to reach a viral audience. This even contributes to the video search function, implementing image recognition technology to allow users to explore content based on visual elements.

In 2023, the platform launched *TikTok* Shop, an e-commerce marketplace available to users straight from the *TikTok* app. This retail feature shortens the customer journey, allowing for seamless purchasing and business marketing. CNNs play a fundamental role in the creation of "shoppable videos" on the site. Using image recognition, the network identifies a particular product within a video and automatically provides a link to the product listing, allowing users to purchase the item straight from ther video [18]. Influencer marketing is a popular strategy for brand promotion on the platform, so CNNs have the capacity to recognize product placements within a user's content and target users who align best.

## 3.2 Split Testing for Personalized User Profiling

Within the content recommendation system, the platform leverages several probability models to assist in personalizing the user experience. For a particular individual on the platform, *TikTok* likely performs continuous split testing to determine what type of content resonates best with the user. The algorithm is constantly adjusted in real time. Based on user retention, engagement rates, and video shares, the model evaluates the success of various versions of the algorithm by calculating the likelihood of performance outcomes. For two given algorithm types $A$ and $B$, split testing determines if there is a statistically significant difference in performance between the variants. A null hypothesis, $H_0$ is defined, which assumes that there is no difference in user engagement between videos recommended by each algorithm. Thus,

$$H_0 : \mu_A = \mu_B \tag{23}$$

for the expected mean values of a particular engagement metric, (i.e. average watch time). The alternative hypothesis $H_1$ identifies that there is a difference in engagement.

$$H_1 : \mu_A \neq \mu_B. \tag{24}$$

Tik Tok runs an experiment to a randomly assigned group of users, designating half of them to view videos recommended by $A$ and the other half to $B$. They will collect data from the groups over a certain time interval and compare the results in order to improve the algorithm. After data collection, the platform calculates a t-test to compare the engagement metrics for each group and determine if a statistically significant difference exists.

$$t = \frac{\overline{x}_A - \overline{x}_B}{\sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}}. \tag{25}$$

The variables are defined as follows:

$\overline{x}_A, \overline{x}_B$ are the mean watch times generated by each respective algorithm

$s_A, s_B$ are the standard deviations of watch times for each group

$n_A, n_B$ are the number of users in each group

Using the result of this formula, the platform identifies the corresponding p-value. If the p-value is less than 0.05, then the null hypothesis is rejected, since there is a statistically significant difference between the engagement metrics of the algorithms. Thus, Tik Tok would conclude that $B$ produces better user engagement. If the p-value is greater than 0.05, there is no significant difference, meaning that $A$ and $B$ perform similarly. This strategy is leveraged to make real-time adjustments to the algorithm, ensuring that users are consistently coming in contact with videos that align best with their preferences. If a significant difference in engagement is calculated, Tik Tok will push the "more successful" algorithm to the user and continue to show them similar content. This process is repetitive, as user preferences change quite rapidly on the platform. Every individual's FYP is entirely unique and everchanging, which is what makes the app so addictive.

## 3.3 Hashtags for Content Categorization

Tik Tok uses hashtags, (words or phrases accompanied by the # symbol), to categorize content on their platform. This assists in making videos more searchable for viewers, contributing to the ease of use and relevance of the app.

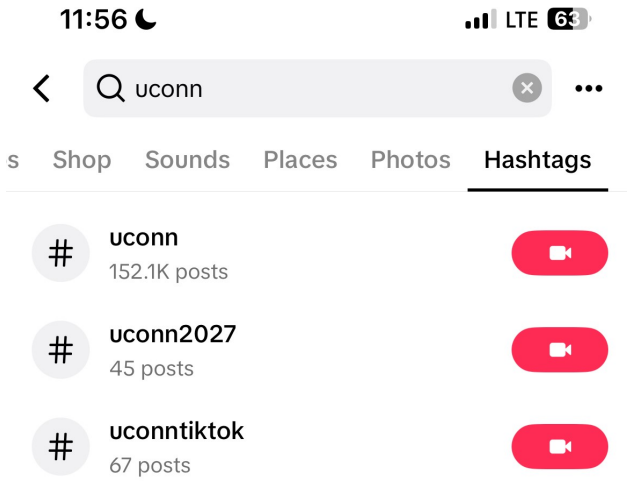
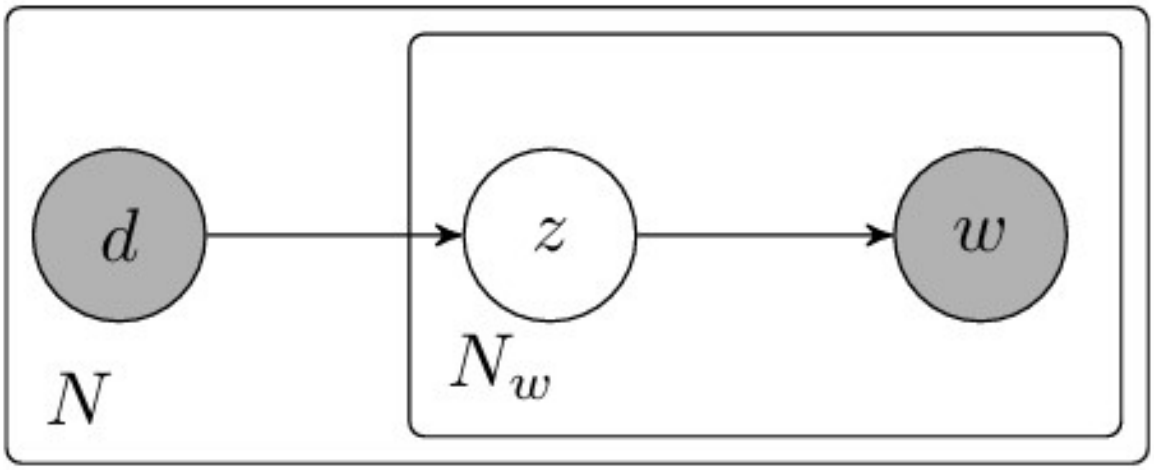



Figure 1: Hashtags on *TikTok* assist in content categorization, trend amplification, and algorithmic discovery. By tagging videos with relevant terms, users increase their visibility. Additionally, the algorithm uses this data to group and recommend content to target audiences.

Probability models are a key component of hashtag systems, as they accurately predict which tags are most related to a given post. Probabilistic Latent Semantic Analysis (PLSA) is a technique primarily used to identify the underlying themes in a post and assign probabilities to the topics based on words used. With this data, the system is able to recommend hashtags associated with that topic. Co-occurence is a key factor in this process. The model detects strong associations between which hashtags are often used together to provide more precise recommendations to users. The structure of PLSA is based on the aspect model, which associates hidden variables (topics/concepts) with observed variables (text). The technique expresses data in the form of 3 variables:

- Documents ($d$)
- Words ($w$)
- Topics ($z$)

**Figure 2: Graphical Model of Variable Filtering**




*Source: Probabilistic Latent Semantic Analysis*

For $N$ documents in a given dataset, $N_w$ represents the number of words in a given document. Each word is associated with a given concept, $z$. Documents and words (shaded) are observed variables, while concepts (unshaded) are latent variables. The generative process associates each topic, $z$ with an observed pair $(d, w)$. The pair is formed by first selection a document $d_n$ with an assigned probability $P(d)$. For each word,$w$, in the document, a topic $z$ is selected from a multinomial conditioned on the probability $P(z|d_n)$. A word is also selected from a multinomial conditioned on the previously chosen topic with probability $P(w|z)$. Using the product rule, we obtain $P(d, w)$:

$$P(d, w) = P(d)P(w|d). \tag{26}$$

$$P(w, d) = \sum_{z \in Z} P(w, z|d). \tag{27}$$

$$= \sum_{z \in Z} P(w|d, z)P(z|d). \tag{28}$$

Given that words and documents are conditionally independent given the topic, we know

that

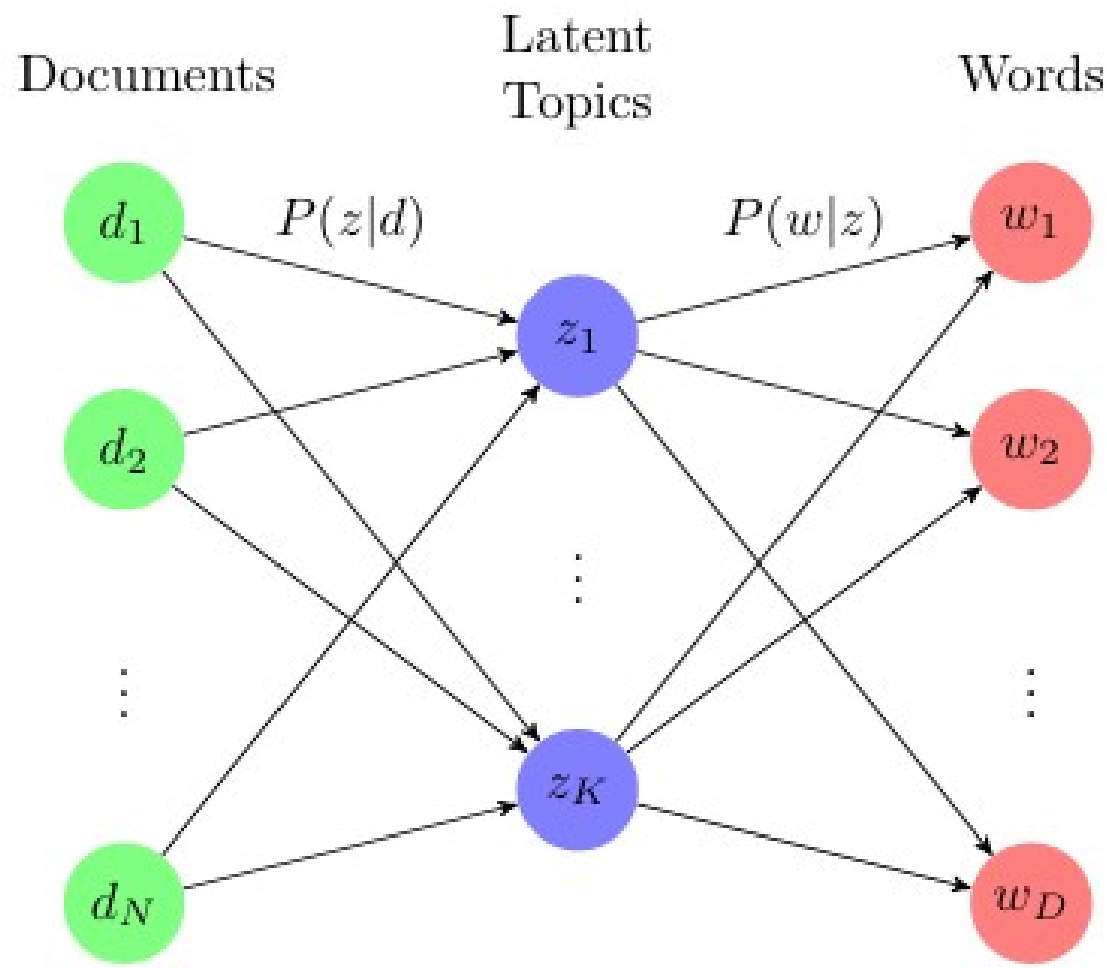
$$P(w, d|z) = P(w|z)P(d|z) = P(w|z).$$

Thus, the predictive probability of the PLSA model can be mathematically represented as:

$$P(w|d) = \sum_{z \in Z} P(w|z)P(z|d).$$

The PLSA model can be represented via the following graphical image:

**Figure 3: PLSA Model Configuration**




*Source: Probabilistic Latent Semantic Analysis [10]*

This image depicts the intermediate layer of topics that link documents and words together. Essentially, each document is represented as a combination of concepts weighted by the probability $P(z|d)$ and each word is associated with a topic with probability $P(w|z)$.

Using likelihood maximization, the model calculates the values that increase the probability of observed word occurrences. This probability, $P(w|d)$, is maximized given the following equation

$$L = \Pi_{(d,w)} P(w|d) = \Pi_d \Pi_w P(w|d)^{n(d,w)}.$$

where $n(d, w)$ is the number of times a word appears in a given document. To optimize this, the following calculation is performed:

$$\mathcal{L} = \log L = \sum_d \sum_w n(d, w) \cdot \log \sum_z P(w|z) P(z|d).$$

The PLSA model, when applied on a set of images from a video, can extract object categories. Using learned objects store within a data class, images are segmented into groups based on local features that belong to each specific object. In the context of video-related content, the "documents" are the images, and the "words" are substituted by visual elements. The training process of PLSA provides the probabilities $P(z|d)$ and $P(w|z)$. Using $P(z|d)$ for each image, the model classifies images as a particular object $k$, where $k$ represents the points at which $P(z|d)$ achieves its maximum. [10]

PLSA provides the algorithm with a convenient process for understanding the relationships between different pieces of content and the usage patterns of various hashtags. By categorizing videos with these tags, users are able to effortlessly find and follow posts related to a specific topic of interest, positively contributing to the user-friendly interface of the platform.

Through such features, *TikTok* cultivates a safe online environment driven by dynamic user profiling and community management. In sustaining user satisfaction and trust, the platform is able to set the stage for continued growth and value generation.

# 4 Producing Creator Value through Content Production Assistance

This section investigates how TikTok produces creator value by offering advanced content production tools powered by complex analysis and AI. TikTok processes and transforms audio signals to help creators sync and enhance their videos with trending sounds. AI-driven filters and effects use computer vision and real-time signal processing to enhance visual storytelling. Together, these tools reduce technical barriers and enable creators to produce engaging, polished videos that are more likely to perform well on the platform.

## 4.1 Complex Analysis for Audio Optimization

Music is a key driver of engagement on *TikTok*, so the application provides creators with many unique audio features that assist in the content production process. One of the most popular digital function is auto-sync, which automatically aligns videos with audio beats. This is likely facilitated by Fast Fourier Transforms, which convert signals into spectral components and collect frequency information on a particular sound clip. Using this information, the app is able to extract beats and patterns within the music as well as enhance noise reduction and pitch shifting technology on the platform.

Fourier transforms convert a function from the time domain to the frequency domain, decomposing each sound into individual pitches and assessing how much of each frequency is present. Each frequency component is a single sinusoidal oscillation, uniquely characterized by their own amplitude and phase.

Let us dive into the formulaic construction of Fourier Transforms [4].

**Definition 2** (Fourier Transform of $f(x)$)**.**

$$F(\omega) = \int_{-\infty}^{\infty} e^{-i\omega x} f(x) dx. \tag{29}$$

In the context of audio, $f(x)$ refers to the input signal/data, while $F(\omega)$ is the signal's spectrum (makeup of frequencies). The variable $f$ is the impulse response of a filter on the input data to produce the output data. Then, $F$ represents the filter's frequency response.

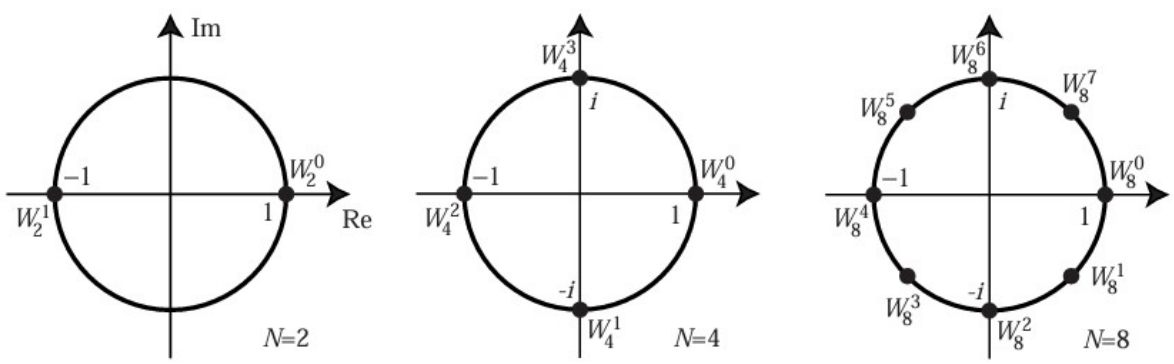
Fourier Transform requires continuous signals for computation, but since most signals are digitized and sampled at various intermittent integrals, Discrete Fourier Transform (DFT) can be implemented to work with sample data. Using this method on periodic, discrete data sets, we can obtain a spectrum list of frequencies and measurement of how much of each frequency exists.

**Definition 3** (Discrete Fourier Transform of $a$)**.** *Define a signal $a_n$ for $n = 0, 1, ...N-1$. Let $a_n = a_{n+jN}$ for all $n$ and $j$. Then the spectrum of $a$ is defined as*

$$A_k = \sum_{n=0}^{N-1} e^{-i\frac{2\pi}{N}kn} a_n. \tag{30}$$

Let $W_N = e^{i\frac{2\pi}{N}}$. For $k = 0...N-1$, $W_N^k$ are considered the *$N$th roots of unity*, since $(W_N^k)^N = 1$ for all $k$. The roots of unity for $N = 2, 4$, and 8 are graphed below:

**Figure 4: Roots of Unity**




*Source: Fast Fourier Transform [4]*

Since the $N$th roots of unity are points on the unit circle every $\frac{2\pi}{N}$ radians apart, the powers of roots of unity are periodic with period $N$. Multiplying by $W_N$ rotates clockwise

by $2\pi$ radians. Thus, $A_k$ is the discrete Fourier Transform of the signal $a_n$. Discrete Fourier Transform is a time-consuming practice, especially for large data sets. For a list of $N$ data points, DFT requires $N^2$ operations ($N$ multiplications and $N$ additions). The Fast Fourier Transform (FFT) Algorithm computes DFT much more quickly and efficiently. This method allows the information to be broken into smaller bits, solved recursively, and then recombined for accurate results. FFT only costs $\mathcal{O}(N \log N)$.

Prior to defining the FFT algorithm, it is important to denote the polynomial $p(x)$ (of degree $n$) that the algorithm is based upon.

$$p(x) = a_0 + a_1x^1 + a_2x^2 + ...a_{n-1}x^{n-1}. \tag{31}$$

We assume $n$ os a power of 2, yielding $\omega$ to be the $n$th root of unity. For efficiency, we can divide the polynomial into odds and evens:

$$p_{even}(x) = a_0 + a_2x^1 + a_4x^2 + a_6x^3...a_{n-2}x^{\frac{n}{2}-1}. \tag{32}$$

$$p_{odd}(x) = a_1 + a_3x^1 + a_5x^2 + a_7x^3...a_{n-1}x^{\frac{n}{2}-1}. \tag{33}$$

With this, we know

$$p(x) = p_{even}(x^2) + p_{odd}(x^2). \tag{34}$$

In separating these polynomials, we evaluate $p_{even}(x)$ and $p_{odd}(x)$ at $\frac{n}{2}$ complex $\frac{n}{2}$th roots of unity. The formal algorithm for FFT is written as follows:

**FFT Algorithm:**

***if (n == 1)***

$\quad$ ***return*** $a_0$

$w \leftarrow e^{\frac{2\pi i}{n}}$

$(e_0, e_1, \ldots, e_{\frac{n}{2}-1}) \leftarrow \boldsymbol{FFT}(\frac{n}{2}, a_0, a_2, \ldots, a_{n-2})$

$(d_0, d_1, \ldots, d_{\frac{n}{2}-1}) \leftarrow \boldsymbol{FFT}(\frac{n}{2}, a_1, a_3, \ldots, a_{n-1})$

***for*** $k = 0$ ***to*** $\frac{n}{2} - 1$

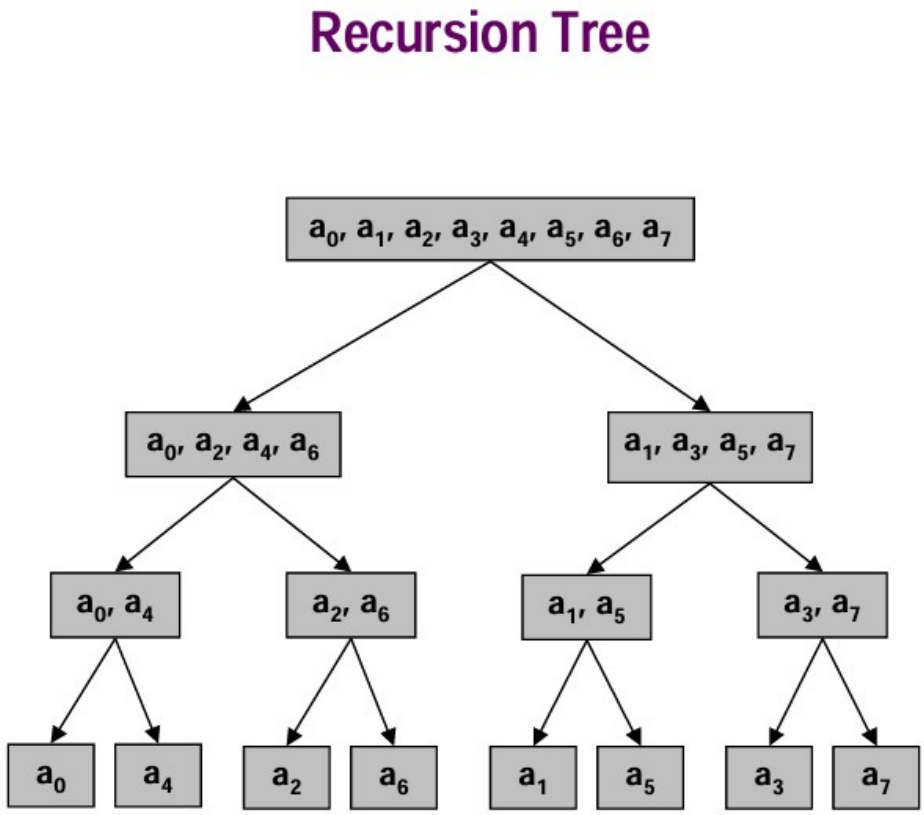
$\quad$ $y_k \leftarrow e_k + w^k d_k$

$\quad$ $y_{k+\frac{n}{2}} \leftarrow e_k - w^k d_k$

***return*** $(y_0, y_1, y_2, \ldots, y_{n-1})$

From this algorithm, the follow recursion tree is produced:

**Figure 5: Recursion Tree**




*Source: [5]*

In proving this algorithm, we showing that $y_k = p(\omega^k)$ for each $k = 0, ..., n-1$, where $\omega$ is the principal $n$th root of unity.

*Proof.* Base Case: At $n = 1$, $\omega = 1$, the algorithm returns

$$y_0 = a_0 = a_0\omega^0. \tag{35}$$

Induction: Assume the algorithm holds true for $\frac{n}{2}$. Let $v$ be the principal $\frac{n}{2}$ root of unity. Then,

$$e_k = p_{even}(v^k) = p_{even}(\omega^{2k}). \tag{36}$$

by (2.). Additionally,

$$d_k = p_{odd}(v^k) = p_{odd}(\omega^{2k}) \tag{37}$$

by (2.). So,

$$y_k = e_k + \omega^k d_k \tag{38}$$

$$= p_{even}(\omega^{2k}) + \omega^k p_{odd}(w^{2k}) \tag{39}$$

$$= p(\omega^k) y_{k+\frac{n}{2}} = e_k - \omega^k d_k \tag{40}$$

$$= p_{even}(\omega^{2k}) - \omega^k p_{odd}(w^{2k}) \tag{41}$$

$$= p_{even}(\omega^{2k}) + \omega^{k+\frac{n}{2}} p_{odd}(\omega^{2k}) p_{even}(\omega^{2k+n} + \omega^{k+\frac{n}{2}} p_o dd(\omega^{2k+n}) p(\omega^{k+\frac{n}{2}}). \tag{42}$$

□

Practically every social media platform relies heavily on FFT for assistance with audio analysis features on their sights. For TikTok, specifically, there are numerous areas with which FFT may come into play. The first is song identification: when you upload a video to the platform, the algorithm analyzes the audio spectrum and compares the resulting frequencies to its database of known songs. In turn, it is able to match up and recognize background music. Additionally, the app provides users with many voice distortion features that are powered by the conversion of domain frequencies. Using FFT, the audio is compartmentalized and modified, boosting or removing components of each frequency based on the applied change. For instance, the popular "Chipmunk voice" feature shifts all of the frequencies upward, producing a high-pitch sounding version of the original audio track. Tik Tok even has ability to reduce background noise on videos; using FFT, it detects a constant frequency (like wind or humming) and subtracts it from the spectrum.

The Sound-Sync is another popular feature of Tik Tok that leverages FFT to create an easy video generating experience for users. The feature matches video footage to the chosen soundtrack, aligning the clips to the beats of the music to boost engagement.

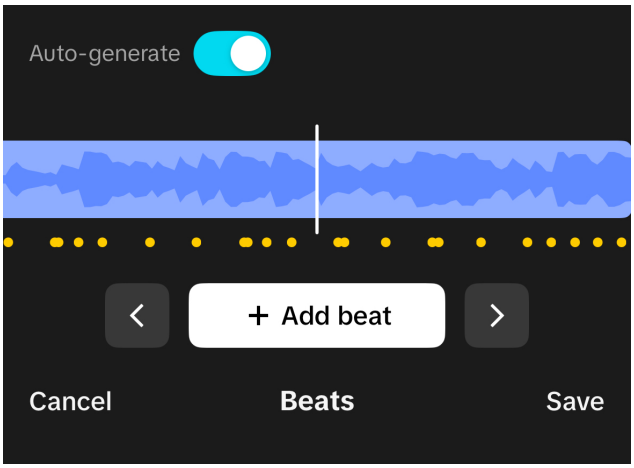



Figure 6: *TikTok*'s Beat Sync feature automatically aligns video clips to the underlying music track by analyzing rhythmic patterns and temporal markers. The yellow dots represent each video frame, which transition according to algorithmic timing and signal processing.

The algorithm breaks down the sound in order to understand the rhythm. It recognizes the tempo of the song as well as the dominant peaks at certain frequencies (i.e bass hits or snares). The frequency breakdown provided by FFT creates a grid of where each beat lands. After uploading a video, Tik Tok detects scene changes and cuts the video to the timing of the beat map, creating a seamlessly edited montage. An example of this is the "Slideshow" feature which prompts the user to choose a series of photos and select a song. The algorithm calculates the number of beats in the audio and evenly splits the photos across those beats, cross fading between them exactly on tempo. Such features aid in accomplishing the app's goal of generating creator and user value. The app essentially does all of the work for you, allowing individuals to create high-quality content with minimal effort required.

## 4.2 Enhancing Creativity with Augmented Reality Features

Augmented Reality filters are a major component of Tik Tok which allow users to overlay digital effects onto their content and engage in interactive, animated experiences. These filters have a tremendous impact on the overall user experience, increasing consumer satisfaction and storytelling ability on the platform. The most popular use of AR filters is for facial transformation, which can change or enhance a user's appearance by adding makeup,

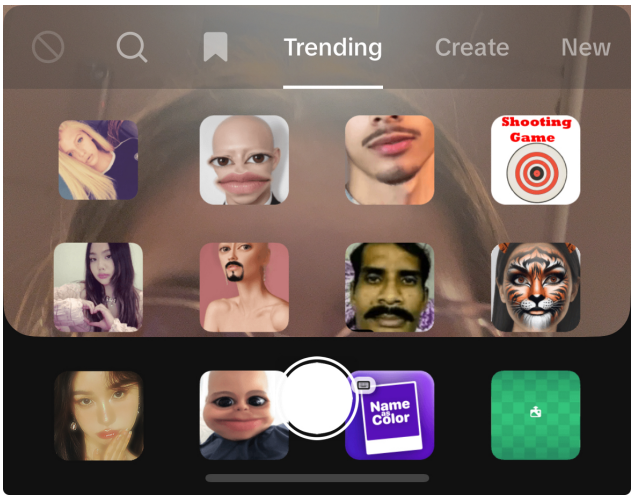



Figure 7: *TikTok*'s face detection and enhancement features. These tools rely on computer vision and machine learning algorithms to identify facial landmarks, apply filters in real time, and personalize user content.

accessories, or character distortions. Using the convolutional layers detected by the network, facial landmarks such as eye shape, nose, and mouth are recognized.The CNN is trained on large datasets which assist in locating these features and therefore accurately placing the virtual elements onto each frame. Temporal tracking is used to align and sync the filters with the movements of each facial feature. The most prominent technique to temporal tracking is optical flow, which estimates the motion of objects between frames based on patterns of brightness [12]. The model observes how the intensity of each pixel has changed across frames, grouping adjacent pixels together to estimate how much a prominent object has moved. So, as an individual blinks or shakes their head, the associated AR filter will remain in place relative to the user's face.

By offering a wide variety of in-app editing and audio features, *TikTok* simplifies the production process for creators. This enhances creative expression on the platform and encourages participation among all users, increasing their visibility on the app and fostering an accessible, creator-driven environment.

# 5 Conclusion

In this thesis, I investigate how *TikTok*'s algorithm, by leveraging key principles of linear algebra, probability, machine learning, and mathematical analysis, generates value for individual users, content creators, and the platform as a whole. At the heart of the operating system is a sophisticated content recommendation engine, which uses high-dimensional vector representations that match users with videos based on viewing habits. Probabilistic models help anticipate user behavior, allowing the platform to display content that maximizes engagement and retention. Machine learning enables content personalization in real time by learning user interactions to shape each user's For You page. The algorithm relies on these computational tools to detect and filter out inappropriate content, contributing to the user-friendliness of the platform. In addition, *TikTok* integrates AI-driven video creation assistance into its user inner face, allowing even the most novice of creators to generate content. Together, these features contribute to the continuous generation of user, creator, and platform value on the app, maximizing its success.

While these mathematical models help drive engagement, they simultaneously promote addictive behaviors and exposure to harmful content due to the human biases that computer systems possess. The public raises many concerns about the fairness and acceptability of using information processing systems to make real-world decisions. Researchers have identified a number of concerns about the ways in which *TikTok* negatively impacts its current generation of users.

The rapid pace of *TikTok*'s short-form content delivery facilitates an environment in which users are constantly stimulated and experience instant gratification. Users have coined the term "doom scrolling" to describe the rabbit hole of excessive watch time they pour into the app due to its addictive nature [15]. Studies indicate that *TikTok* is impacting the attention spans of today's youth, presenting challenges in their ability to concentrate for prolonged periods of time. Excessive use of the app is directly linked to academic challenges; students who spend time on the platform often struggle with information retention. *TikTok* has ha-

bituated this generation's desire for immediate satisfaction and users' working memory is slowly weakening as a result [3].

Mental health is also a major worry, as speculation has risen that the algorithm tends to amplify depressive content. The system allegedly pushes content related to mental health challenges, exposing children and young adults to concepts surrounding depression, anxiety, and self-harm in pursuit of capturing their attention [16]. In a 2023 research study performed by Amnesty International, after 6 hours of scrolling on the platform, almost 1 out of every 2 videos was mental-health related. The app's engagement-focused business model potentially poses serious health risks to adolescent users [1].

Perhaps the largest issue regarding *TikTok*'s algorithm is national security. The Chinese government's control of ByteDance has amassed great congressional concern in the US. In fear that China could obtain access to American users' data, a de jure nationwide ban was placed on the platform [11]. While President Trump signed an executive order delaying the enforcement of this law, *TikTok*'s data collection practices are still a major topic of conversation. Although components like browsing history and location data are pretty universal to the operation of most social media sites, the foreign ownership of *TikTok* poses potential cybersecurity risks.

In order to promote a healthier experience for users, *TikTok* should consider reducing the algorithm's reliance on dopamine-triggering content and make platform controls more transparent to users. Viewers should be able to see why videos are recommended to them and have the ability to adjust their preferences accordingly. While *TikTok* keeps this information predominantly confidential, applications of machine-based learning models and algorithmic methods reveal insightful details about the ways in which our data is collected and processed. Developing an understanding of how social media algorithms function is critical, as these technologies occupy so much of our time and influence the information that we are exposed to. In recognizing the strength of these recommender systems, we can better manage our screen times and how we allocate our attention.

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